

SCHOOL OF SCIENCE AND ENGINEERING

THE MODELING OF A REGENERATIVE BRAKING SYSTEM USING BLDC MOTOR IN AN ELECTRIC VEHICLE

Capstone Design – EGR 4402

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Capstone Report

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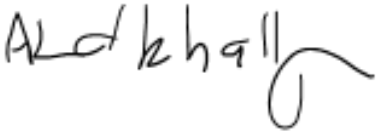
"I, Ikram Rachak, have applied ethics to the design process by using different scholarly sources and citing them, and in the selection of the final proposed design. I have held the safety of the public to be paramount and have addressed this in the presented design wherever may be applicable."

Capstone Student



Approved by the Supervisors

Dr. Ahmed Khallaayoun:



Mr. Rachid Lghoul



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LIST OF ACRONYMS AND ABBREVIATIONS:

KERS: Kinetic Energy Recovery Systems

RBS: Regenerative Braking Systems

BLDC: Brushless Direct Circuit

BLDCM: Brushless Direct Circuit Motor

EMF: Electric and Magnetic Field

BEMF: Electric and Magnetic Field

DC: Direct Circuit

AC: Analog Circuit

PM: Permanent Magnet

PWM: Pulse Width Modulation

PI: Proportional Integral

UBB: Universal Bridge Block

MOSFET: Metal–Oxide–Semiconductor Field-Effect Transistor

Ah : Amp hours

T: Torque (Nm)

Te: Electromagnetic torque

ω: Angular Speed

I: Current through the windings (ampere)

K_T: Torque constant

V: Voltage (volt)

ABSTRACT

One of the issues that the majority of the electric vehicle industry is currently dealing with is the significant amount of energy that is lost as heat during braking operations. To reduce this wasted energy, we can use a regenerative braking system. Regenerative braking converts kinetic energy lost during braking into electric energy. In this project, “The Modeling of a Regenerative Braking System Using BLDCM in an Electric Vehicle”, different ways to recover and store energy in an electric vehicle were presented. A regenerative braking system that uses the BLDC motor was built in Simulink and was able to recover 0.003 % of the battery charge in each brake operation. The brushless DC motor is used because of its high efficiency, ideal torque and speed, and low maintenance costs.

INTRODUCTION

According to the US Environmental Protection Agency, "Internal Combustion Engine cars contribute between 40 to 50 percent of ozone, a80 to 90 percent of carbon monoxide, and 50 to 60 percent of air toxins that are found in urban areas"[1]. With the increase in environmental pollution, electric vehicles are one of the alternatives to these vehicles. Electric vehicles use either AC or DC electric motors. Recently, motors like brushless motors, switched reluctance motors, and motors based on PMS are used in EVs as they yield a higher efficiency. One of the key features of electric vehicles is regenerative energy technology. According to one study, one third (20 to 25%) of energy is lost during decelerating [1]. Regenerative braking systems enable the vehicle to recover a large percentage of this lost energy. That is to say, this is an energy recovery technology that has the potential to significantly reduce the energy consumption of electric vehicles, particularly on urban routes. Regenerative braking systems' performance varies depending on different criteria like the motor used, the mass of the car, speed, torque, temperature, and so on. This project aims to build a regenerative braking system that can recover energy and transform it into electricity while considering the speed of the vehicle and the torque of braking.

1) Objective:

The purpose of this project is to explore different methods to recover and store energy as well as modeling a regenerative braking system that recovers power in each brake operation with the aim of working on improving its efficiency using pulse width modulation later on. To build the model, MATLAB Simulink is used. The complete system will be explained and tested and the results will be shown.

2) Motivation:

Electric vehicles are becoming more popular as a viable alternative to internal combustion engines that do not rely on renewable resources. The use of fossil fuels is rising at a rapid rate leaving a larger carbon footprint and contributing to global warming and pollution, thus, alternative energy sources should be used. Electric cars are one of the best solutions that contribute to lowering the carbon footprint and decreasing pollution. Hereby, the electric vehicle should use the most updated braking system. Regenerative braking systems use renewable energy sources to perform the same task as traditional brakes along with improving electric vehicle energy efficiency. Therefore, they should be installed in automobiles. Using these systems in electric vehicles will not only increase sales but also decrease pollution. The country, especially urban areas, will inherently benefit from the introduction of EVs. The air quality within urban areas will be significantly improved.

3) Methodology:

To understand regenerative braking systems, I started with a chapter about their history. Then the second chapter explains the concept of these systems.

The next chapter: Modeling and simulation, is divided into three parts. In the first part, the concept of the BLDC motor is explained. In the second part, a general block diagram of the simulated regenerative braking system will be presented. The BLDC machine will operate in two modes: the first is as a generator and the second is as a motor. During the motion of the electric vehicle, energy is supplied by a battery. In this project, the current flows from the battery to the BLDC machine. This path is indicated clearly by an orange line in Figure 3.2.1. In the reversed path, the regenerative braking mode, where the battery gets charged from the BLDC generator, the path is indicated by a green line as shown in figure 3.2.1. In the third part, a simulation of the system using Simulink will be represented. Finally, in the last part, the results will be analyzed and discussed.

Ch1: LITERATURE REVIEW

1) History:

The Flybird

The flybrid represented in figure 1.1 is the first kinetic energy recovery system. This system weighs 24 Kg with a capacity of 400 kJ per lap. It gives the driver an extra 80hp over a period of 6.67 seconds (a maximum power boost of 60 kW). The flywheel used in this system weighs 5.0 kg and can rotate up to 64, 5000rpm with a maximum torque of 18 Nm.[2]

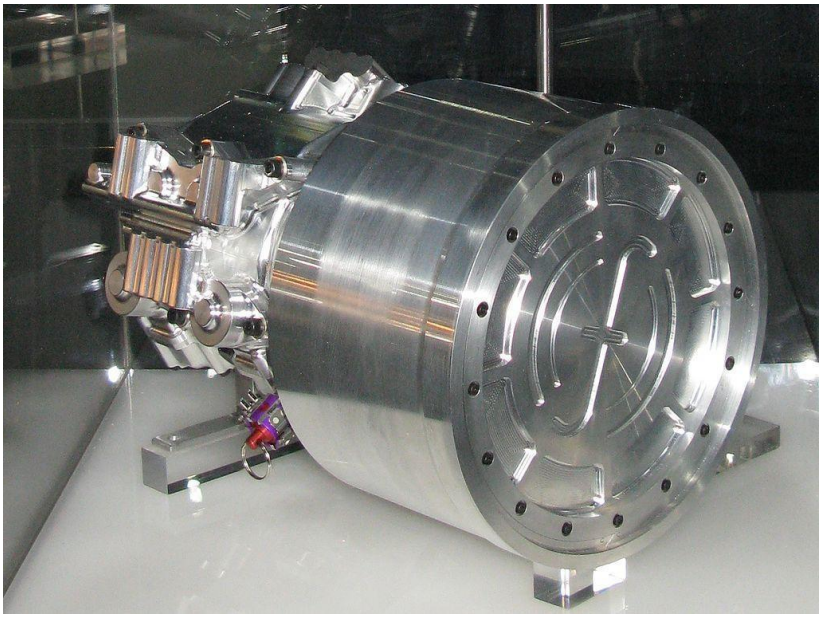


Figure 1.1: The Flybrid Kinetic Energy Recovery System [2]

Formula One

In 2009, the FIA(Fédération Internationale de l'Automobile) approved using 60 KW kERS in the regulations of Formula One. Since Formula One supports eco-friendly solutions, in 2008, participants started using and testing a system that can store energy mechanically or electrically(like in a battery or in a capacitor). While testing KERS, two minor incidents were reported. The first occurred in July when the battery of the Red Bull racing team caused a fire

scare. The other incident happened when a BMW mechanic was a victim of an electric shock when he was in contact with the motor in question. At that time, four teams used these systems including Renault, Ferrari, McLaren, and BMW. Vodafone McLaren Mercedes was the first group that used a KERS-equipped car and won a Formula One in July, 2009. For the second KERS car, it was the fifth. Racer Lewis Hamilton was the first participant to take 1st place with the KERS car. One month later, the Ferrari racer won the Belgian Grand prize with a KERS-equipped car [3].

Car Makers

Many automakers including Honda are conducting research on KERS. Peugeot Sport revealed in 2008 its KERS racing car which represents an electric variant car (Figure 1.2 represents this hybrid electric car). The company intended to race the car in Le Mans Series 2009 season, despite the fact that the car was not able to score at least minimum championship points. Toyota used a supercapacitor to recover energy on its racing car which enabled the team to win the 24 hours Tokachi prize. Regenerative braking was used also by BMW on E90 3 series and on some current models such as the F255 Series. [4]



Figure 1.2: Le Mans Series 2009 [25]

2) Literature Review:

There are different types of braking systems that are used in electric vehicles. Traditional Brakes, for instance, are used in some automobiles. Traditional brakes waste a significant amount of energy since the braking pads heat up. There is another type of braking system which is the anti-lock brakes that use non-continuous braking patterns in order to slow the car. This technology is more efficient and produces better results than traditional brake methods. Another type of effective braking technology to consider is the regenerative braking system. This type uses electric motors. When the driver of an electric vehicle presses the brake pedal, the system inverts the motor's rotational direction, providing torque in the opposite direction as the motion direction. When braking or decelerating, the engine acts as a generator, delivering energy back to the battery while decelerating. These power electronic devices can recharge the battery and aid in decelerating. Formula One motorsport cars have this feature

such that the vehicle recovers the energy lost during braking and stores it in the battery for quick acceleration. This feature has a different name 'KERS' (Kinetic Energy Recovery System).[3]

Regenerative braking systems on the other hand cannot be used completely as they cannot stop the vehicle completely.

There are some other braking systems like dynamic braking and plugging braking. Dynamic braking is frequently avoided due to heat loss and inefficiency. Although plugging braking is fast and efficient, it is not suitable for power generation or electric or hybrid vehicles. Typically found in fast-moving lifts and excavators.

In paper [5] the RBS proposed was able to recover part of the kinetic energy generated as heat during braking. The system can also operate at high temperatures with more efficiency than traditional braking systems. More research is needed to create a better system that recovers more energy. All these systems can benefit moving vehicles by recovering energy that is lost during the braking procedure.

In [24], the authors presented a test bench to test the performance of a brushless DC motor in a regenerative braking system. The project raises engineers' awareness of energy efficiency. This paper focuses on the high efficiency of regenerative braking systems at higher speeds. This technology used in the project contributes to creating a sustainable future in an energy-efficient world [6].

"Siddharth K Sheladia, Karan K Patel, V raj D Savalia, Rutvik G Savaliya, ‘A Review on Regenerative Braking Methodology in Electric Vehicle’, International Journal of Creative Research Thoughts, Volume 6, Issue 1 (2018). 2320-2882." [25]

The RBS used is able to recover from 5% up to 8% of waste energy. Power electronic devices like ultracapacitors, DC-DC converters (chopper), and batteries have been added to the systems. Ultra-capacitors, which aid in improving the car's transient state when starting up, supply smooth charging characteristics, and improve the overall performance of an electric vehicle’s system. Choppers aid in the maintenance of power management in RBS by boosting the vehicle’s acceleration. Flywheels are used in order to improve energy recovery via automotive wheels.[25]

Ch 2: CONCEPTUAL FRAMEWORK

1) REGENERATIVE BRAKING CONCEPT:

Regenerative braking consists of a mechanical method that enables us to recover electric energy from kinetic energy. Electric regenerative braking is more efficient when the electric vehicle stops for a short period of time and then moves again like in traffic jams and traffic lights. Regular braking systems work as a result of the friction between the wheels and the road, and between the green pads and the rotors. According to physics laws, we cannot recover all the lost kinetic energy, but a significant amount of KE can be converted and stored in a battery or a supercapacitor.

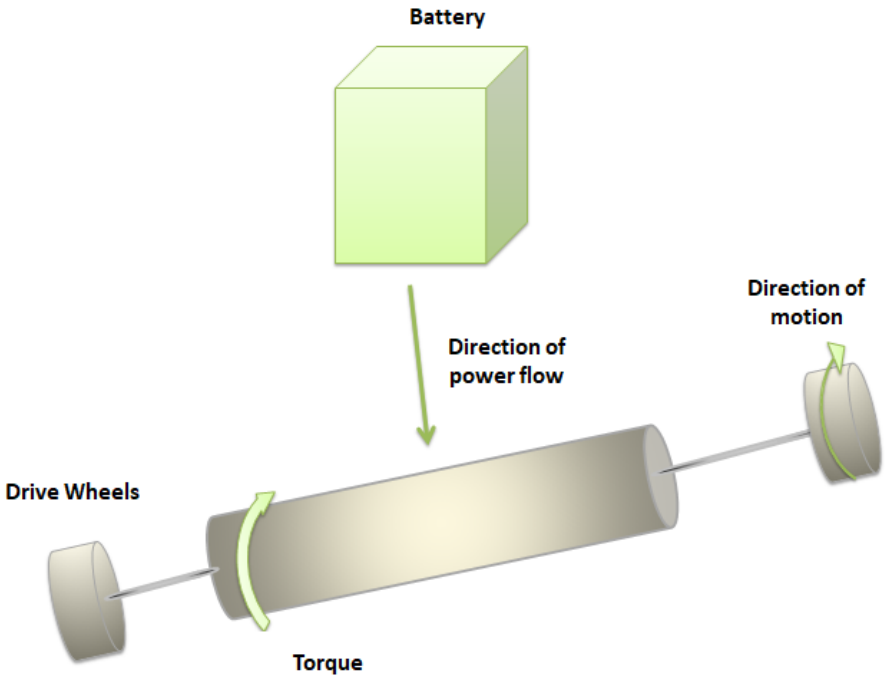


Figure 2.1: When the machine works as a motor

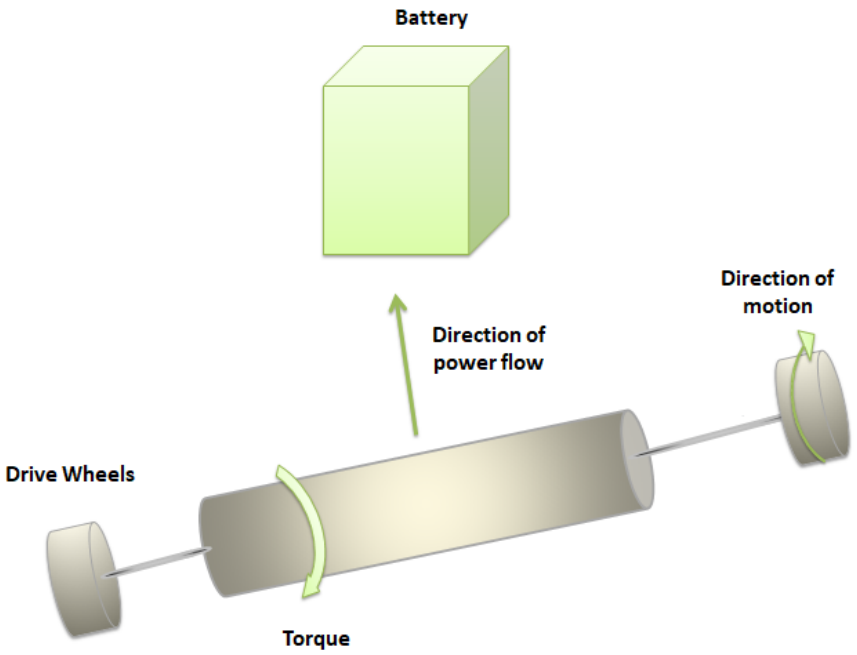


Figure 2.2: When the machine works as a Generator

The electric motor is used to slow the electric vehicle down in RBSs. (Figure 2.1). In this case, torque and motion have the same direction. On the other hand, when the driver presses the brake pedal, the electric motor works as a generator. Torque and motion will have opposite directions leading current to flow in the opposite direction since current and torque are proportionally related. Thus, current will be flowing back to the battery. In this case, the electric motor acts as a generator (Figure 2.2) [7].

2) Equation of Motion:

The equation of motion is expressed as the following: (vehicle's longitudinal motion)

$$m\ddot{x} = F_T - F_{TR} \quad (1)$$

$$F_{TR} = F_T - F_{TR} \quad (2)$$

Where:

F_{TR} : Total Resistance

Fr : Rolling Resistance

F_a : Aerodynamic Force

F_g : Gradient Force

The rolling resistance is a result of the torque and friction that is applied on the tire when the vehicle is in motion.

$$F_r = F_{r,f} + F_{r,r} \quad (3)$$

$$F_{r,n} = \mu_n m_n g \quad (4)$$

Where:

$F_{r,n}$: The Rolling Resistance Force at Wheel n

μ_n : The Coefficient of Rolling Resistance at Wheel n

m_n : The Vehicle Mass at wheel n

g : The gravitational acceleration constant

During motion, an aerodynamic force (F_a) acts on the vehicle due to air contact. The resistance created depends on different factors including the vehicle's shape, speed, and front area.

$$F_a = \frac{1}{2} \rho_a C_d V^2 \quad (5)$$

Where:

ρ_a : Air's density

C_d : The aerodynamic drag coefficient

A : The frontal area of the vehicle

V : Vehicle's Speed.

During acceleration:

$$F_T = \frac{\tau_m}{r_w} \qquad (6)$$

During brake:

$$F_T = -\frac{\tau_r}{r_w} \qquad (7)$$

τ_m : The torque of the motor

τ_r : The resistance torque of the motor.

The traction force can generate thrust force or resist force depending on the control strategies used. The motor produces both torques: driving (τ_m) and resistance torque(τ_r) [9].

3) Types of Regenerative Braking:

To make a regenerative braking system, many methods can be used to convert energy including flywheels, electromagnetics, springs, and hydraulics. There are many types of regenerative braking systems with different energy storage or conversion methods resulting in varying efficiency.

a) Flywheels:

The Flywheel Regenerative Braking System (figure 2.3) collects kinetic energy from the vehicle and uses it to spin a flywheel that is connected via a transmission and gearbox to the driveshaft. The spinning flywheel will then supply torque to the driveshaft, resulting in increased power for the vehicle.[11]

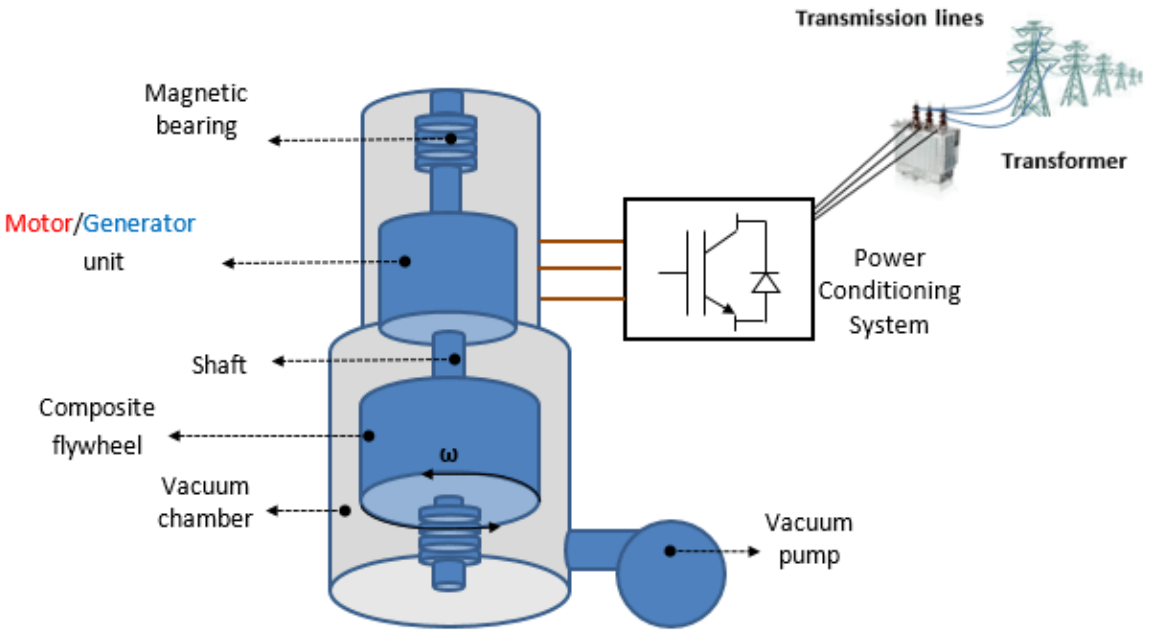


Figure 2.3: Schematic diagram of flywheel energy storage system[11]

b) Electromagnetics:

The vehicle's driveshaft is linked to an electric generator that restricts the drive shaft rotation using magnetic fields, which slows the vehicle and generates electricity in an electromagnetic system (Figure 2.4). For hybrid and full EVs, the recovered energy is sent back to the battery. In conventional vehicles, a battery is used to electrify the electronics in the car, or it might be stored and used later to give the vehicle a power boost.

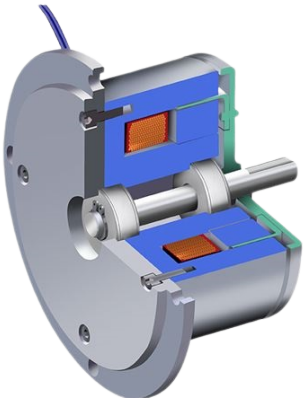


Figure 2.4: Electromagnetic Brake System [12]

c) Springs:

Spring-loaded regenerative braking systems (Figure 2.5) are commonly found on vehicles that are human powered including bicycles or wheelchairs. During braking, a spring is used to store energy as elastic potential energy in the spring RBS. The system that powers wind-up radios, timers, and some torches are essentially the same [13].

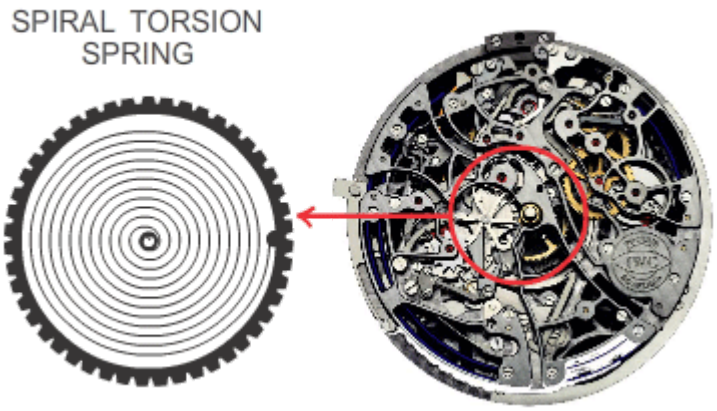


Figure 2.5: Spring Energy Storage System [13]

d) Hydraulics:

The Hydraulic RBS (Figure 2.6) slows the vehicle speed by generating electricity used to compress a fluid. The fluid used frequently in hydraulics is Nitrogen gas. Hydraulic RBSs have a longer energy storage capability compared to other systems because the compressed fluid does not dissipate energy over time. On the other hand, this type of storing energy is a time-consuming process that severely limits the power of the hydraulic RBS[14].

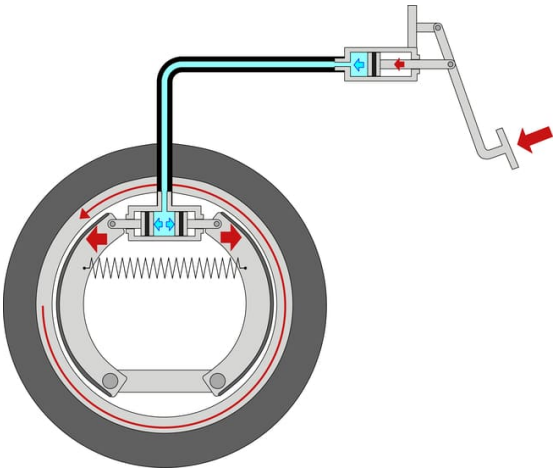


Figure 2.6: Hydraulic Braking System [14]

e) Electromagnetic-Flywheel:

The electromagnetic flywheel RB is a hybrid of the electromagnetic and flywheel RBSs. It uses the same fundamental power conversion techniques that the electromagnetic system uses. In this type, flywheels are used to store energy instead of batteries. They function as a mechanical battery that stores and recovers electric energy. The electric flywheel RBS is considered to be the morecost-effective electricity storage technology due to the longer life of flywheels compared to lithium-ion batteries [14].

4) Advantages & Disadvantages:

Regenerative braking systems have a variety of benefits such that it extends the battery charge, improves the efficiency of braking, and decreases brake wear. The main quality of this technology is that it is eco-friendly: it reduces pollution.

On the other hand, regenerative braking systems have some disadvantages. The system cannot recover all wasted energy while decelerating from high speeds and sometimes cannot stop the vehicle completely. Thus, we cannot use the regenerative braking system as the main braking system of a vehicle.[16]

Ch 3: Modeling & Simulation

1) Working Principle of BLDC motor:

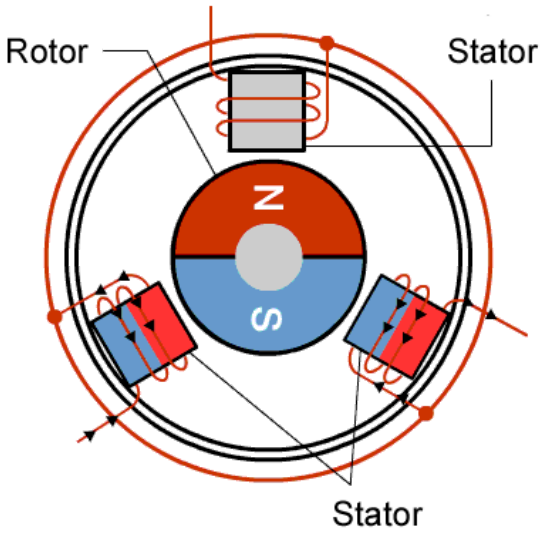


Figure 3.1.1: The BLDC Motor Concept[17]

The BLDC motor is frequently used in electric vehicles as it has a higher peak efficiency and no heat loss due to friction. They are quieter and last longer than some other types, but more expensive.

Brushless DC motors are direct current machines that are ordinarily made up of two parts as figure 3.1.1 shows: a rotor and a stator. For the rotor, it is the permanent magnet that represents the rotating part of the BLDC motor. The stator represents the fixed part of the field winding where a stationary magnetic field with a fixed amplitude is produced. For the coils, they are fixed to the stator. Rotation is accomplished by changing the direction of the magnetic fields generated after inducing current and can be controlled by adjusting the current's direction and magnitude.

In this project, I will use the permanent magnet brushless DC motor that has trapezoidal back-emf waveforms. This motor requires six rotor positions in each electrical revolution to

achieve phase synchronization and needed torque. The position information can be acquired using Hall sensors that are located in the BLDM [17].

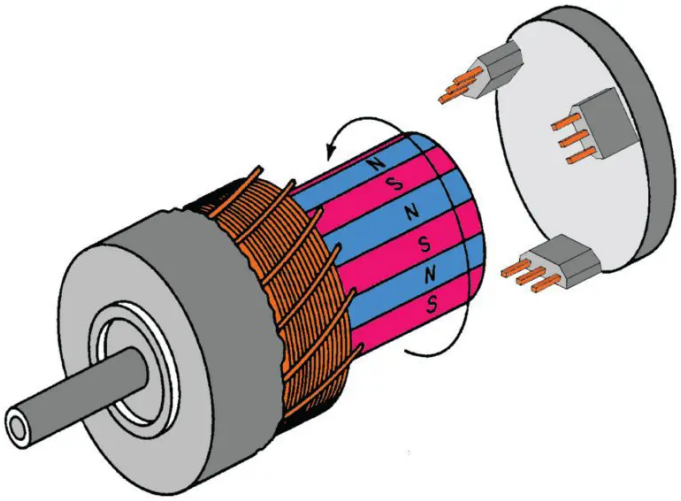


Figure 3.1.2: Hall sensors that are inside the BLDC motor [18]

In order to connect the BLDC motor to the battery, I need to use a power converter:

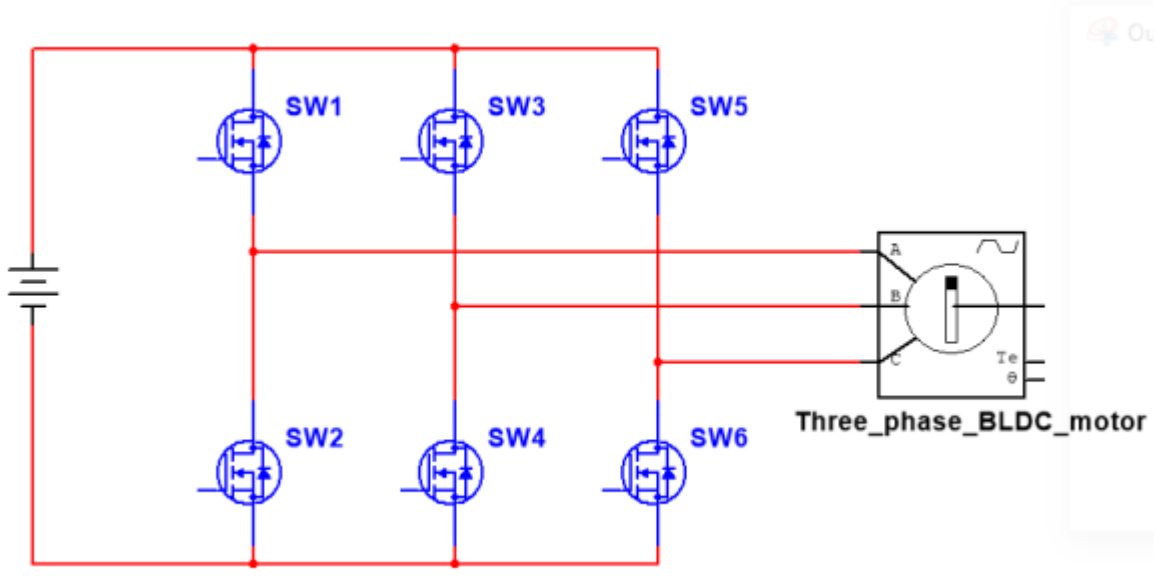


Figure 3.1.3: Six pulse power converter for BLDC motor [19]

This figure shows the power converter that will be used in our project.

In the acceleration phase:

The current going through the motor battery is reversed and directed into the supply battery during acceleration, resulting in regenerative braking. Independent switching using pulse width modulation is an effective method to better control the brakes.

In the regenerative braking phase:

The current in windings follows the direction of the torque which enables us to recover energy and recharge the battery.

The force applied to brakes is not uniform, hence we will use pulse width modulation to control the system and to recover the maximum energy possible.

From the equation:

$$T = I * K_T \qquad (8)\ [20]$$

Where:

I = Windings Current

K_T = Torque constant (specific to the motor)

Since torque is proportional to current, a current controller can be used in applications where the motor must operate at a constant torque regardless of the speed. In our case, the controller used is a PI controller. The duty cycle of the PWM pulses will be determined by the output of the current controller. The entire setup can do regenerative braking if the current control mechanism of PWM is used.

2) RBS using BLDC Diagram:

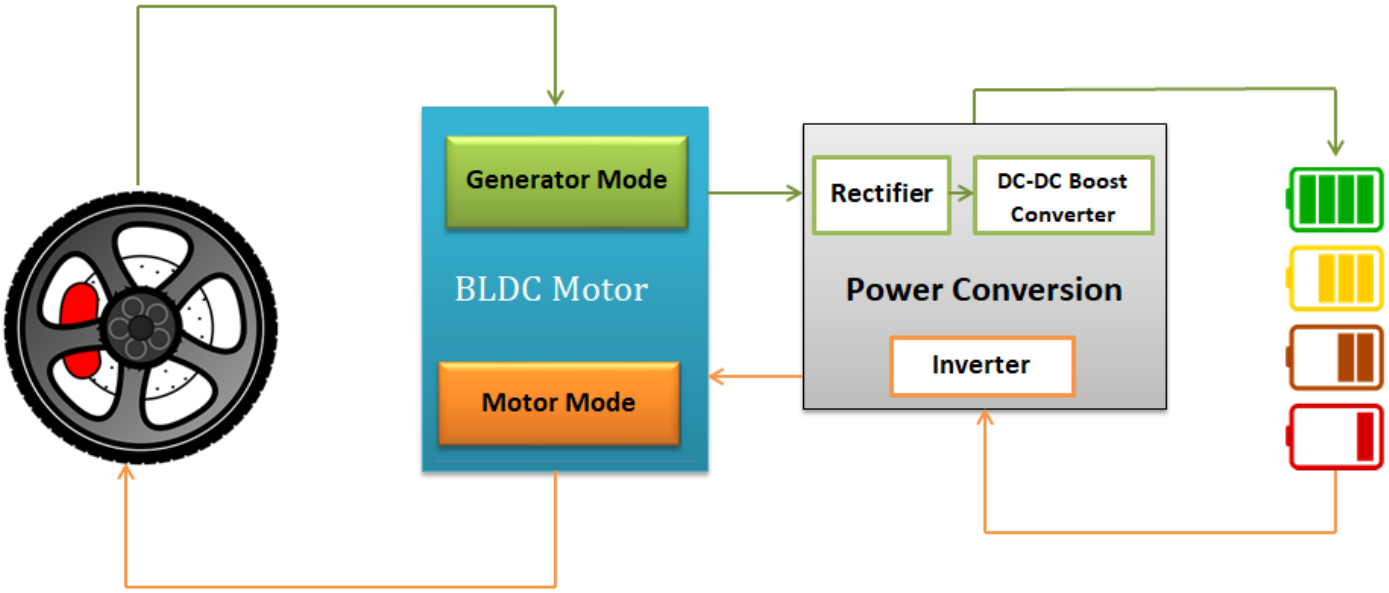


Figure 3.2.1: RBS Block Diagram

Figure 3.2.1 represents how the system built works in both operations:

1st Operation: Acceleration

The vehicle accelerates slowly until it reaches a stable speed based on the input torque. There will be some current spikes present when the motor starts working then they decrease. The current will be supplied to the motor here from the battery, and it will be positive current according to the general sign convention for current measurement. Thereby, the forward motion will be achieved.

2nd Operation: Braking

During forward motion, when the accelerator pedal is depressed, the motor will provide reverse torque to the vehicle as it slows down. The regenerative braking system will be activated, resulting in the charging of the battery. Because the direction has been reversed, the current will be feedback to the power source with a negative sign. In other words, the current

will go back to the battery (generator). The motor will accelerate for a certain period of time by external input and then the regenerative system will require the braking input. The braking is managed by a controller and hall effect sensors for switching of inverter/ motor controller.

3) RBS using BLDC Simulink Model:

The virtual prototype of the system was created in Matlab Simulink R2021b. More details of the model and submodels are shown in the appendix, along other physical constants of the system. I will use a BLDC motor to build a regenerative braking system along with some other power electronics components. The three-phase BLDCM model is constructed using a PI controller for running and braking mode processes under trapezoidal back EMF in this study. For the system, metrics such as current, speed, Back EMF, and torque are examined.

➢ The battery:

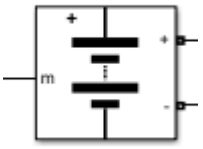


Figure 3.3.1: The battery used

The battery is the most important component of electric vehicles. It ensures storing and supplying electric energy to the whole system. During regenerative braking, it represents the storage of the energy recovered. Since the battery I am using will not work only as a voltage source but also as energy storage, we need to provide more properties like the capacitance that represents the rating of the battery and more.

For this simulation, standard values were used as the specifications of some components [21]:

Lithium Battery

Vsource = 48V

The initial state of charge = 100%

The battery response time = 3 S

Rated Capacity = 80 Amp-hours (Ah)

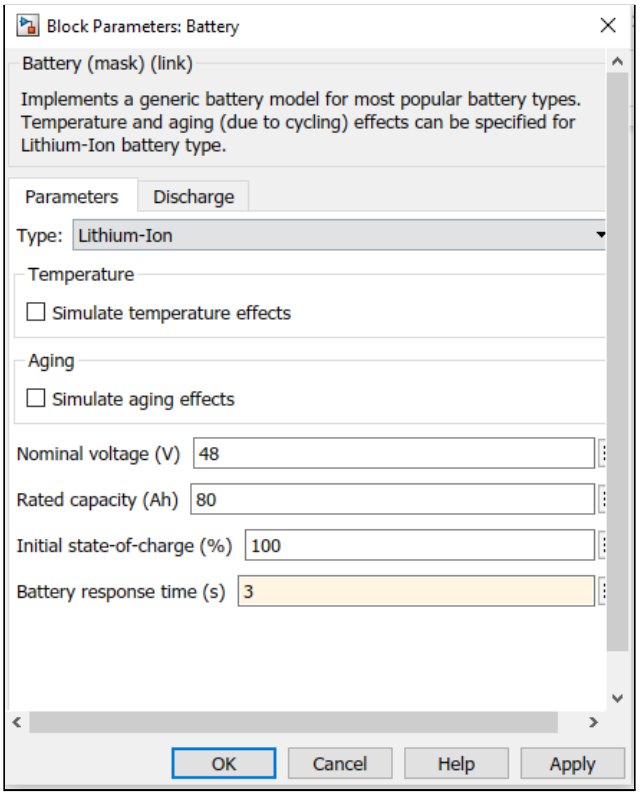


Figure 3.3.2: Battery Parameters

For the discharge parameters, here are the standard values used:

Maximum Capacity = 84 Ah

Cut Off Voltage = 36 V

Discharge current = 27 A

Resistance = 0.006 Ohms

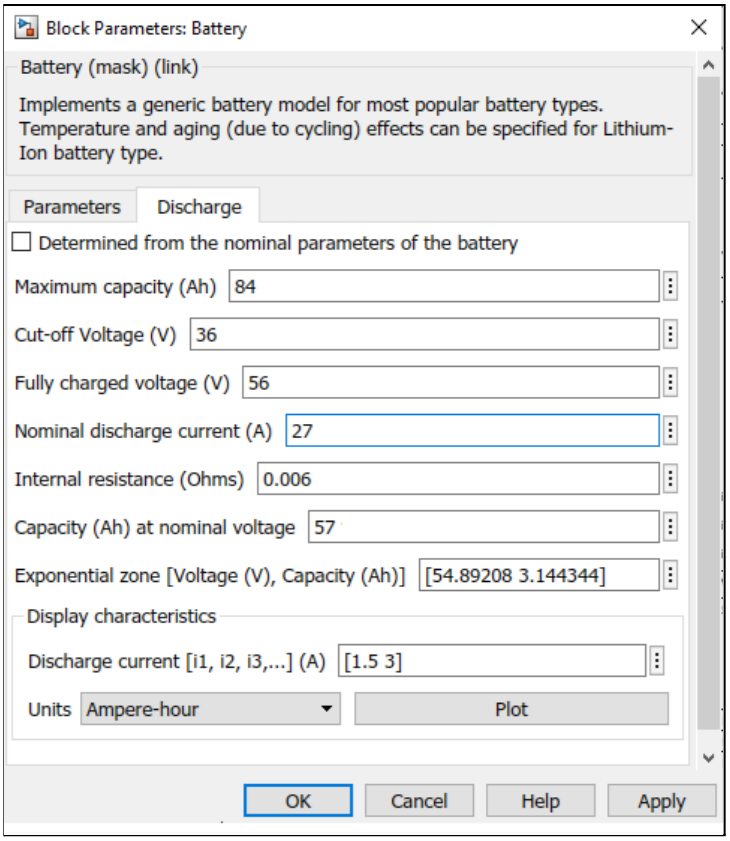


Figure 3.3.3: Battery Discharge Parameters

➢ The Universal Bridge:

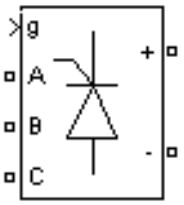


Figure 3.3.4: The Universal Bridge Block

For the BLDC motor, I will use the universal bridge block. From the dialog box, I should set the MOSFET diode as a switching block. The UBB uses a three-phase universal power converter made up of six power switches that are connected in a bridge configuration. These switches allow fast switching.

The main principle of regenerative braking is to change the switching scheme to send the recovered energy back to the battery. The circuit operation consists of two basic voltage boosting states: storing phase and energizing phase. During regenerative braking in the energy-storing state, the bottom MOSFETs, S4, S6, and S2, are switched to form a braking current waveform to provide proper braking performance. The motor driver converter rectifies and boosts a nearly trapezoidal EMF to charge the battery. When a bottom switch is turned off, the power, i.e., current, is transferred to the battery via the top and bottom diodes.

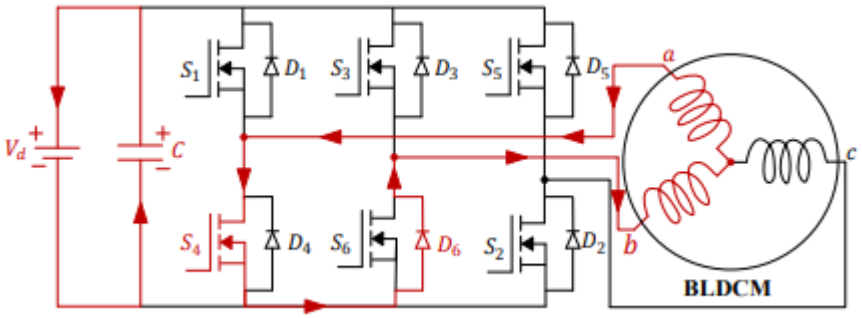


Figure 3.3.5: (a) Storing Mode

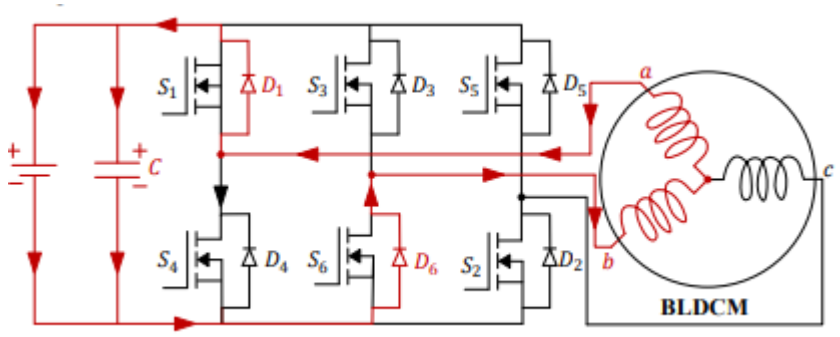


Figure 3.3.5: (b) Energizing Mode[22]

The operation has two states of basic voltage boosting operation: storing and energizing mode. The back EMF induced in the BLDCM during braking is insufficient to store energy because I need to apply a higher voltage than the battery's voltage. In this case, the voltage of the BLDCM during regenerative braking should be amplified by the UBB to achieve the minimum voltage level to store energy. [23]

$$\frac{V_o}{V_{in}} = \frac{1}{1-D} \qquad (9)$$

Where: Vo = The output voltage

Vi = The input voltage

D = The duty cycle in which the switch remains in position.

➢ The Motor:

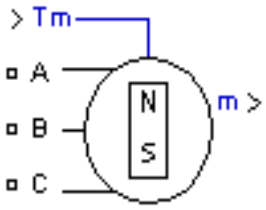


Figure 3.3.6: Permanent Magnet Synchronous Machine Block

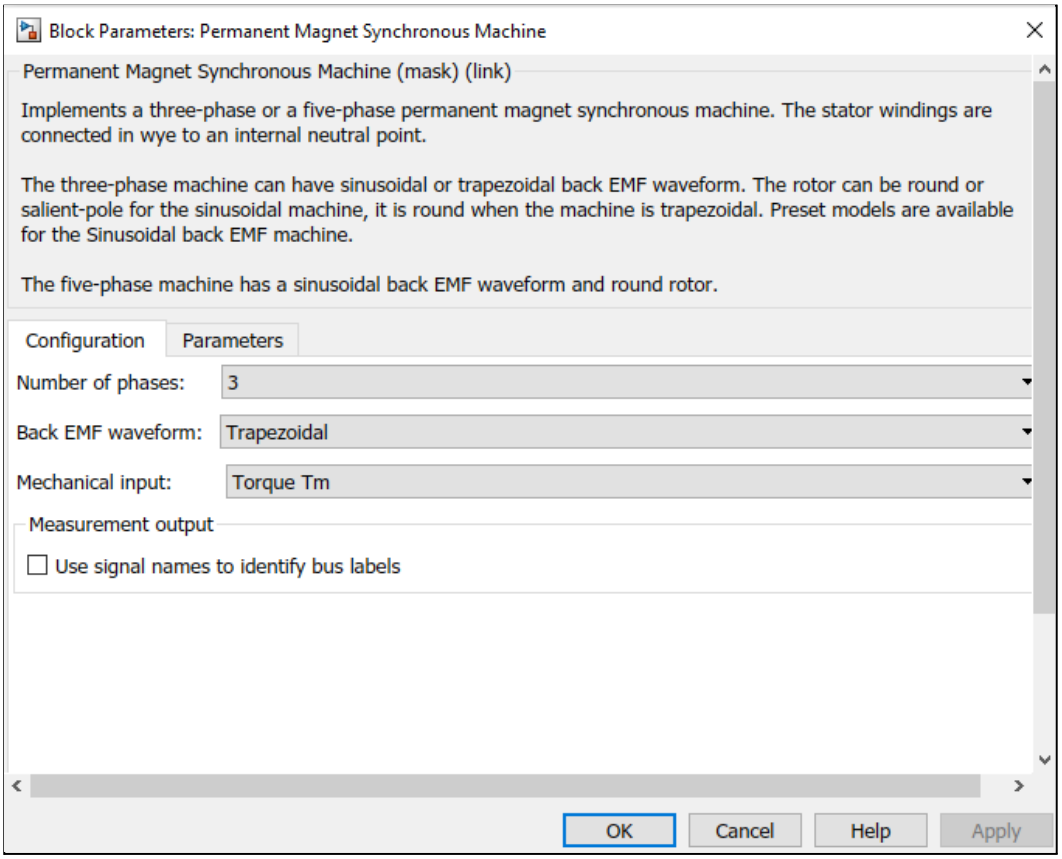


Figure 3.3.7: Permanent Magnet Synchronous Machine Parameters

In MATLAB Simulink, the permanent magnet synchronous machine block is used as a BLDC motor. Considering the two types of this machine: one requires the torque as input while the other wants angular velocity input. Both the torque and the angular velocity data represent the torque or the angular velocity generated on the shaft. Since in this project I used a constant load, torque data is preferred. This block has one input which is the mechanical torque. To enable it, mechanical input should be set to Torque Tm. It has one output port that is a 13-element vector. The three other ports: A, B, and C represent the specialized electrical conserving ports that are associated with the electrical terminals of phases A, B, and C. The motor will be connected to three bus selectors. The first bus selector gives the output in scope and the last one is connected to the input of the back emf of the hall system to generate emf signals.

➢ The Controller:

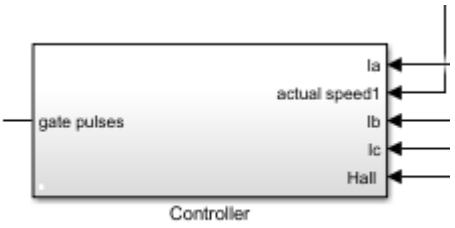


Figure 3.3.8: The controller block

This block represents the controller; it contains a hall sensor, a speed controller, and a current controller. It detects the hall signal, the stator current, the rated speed, etc. The measurements will be made with the help of the signal taken from the m-end of the motor and the scope blocks. The following model shows the submodel that is inside the controller:

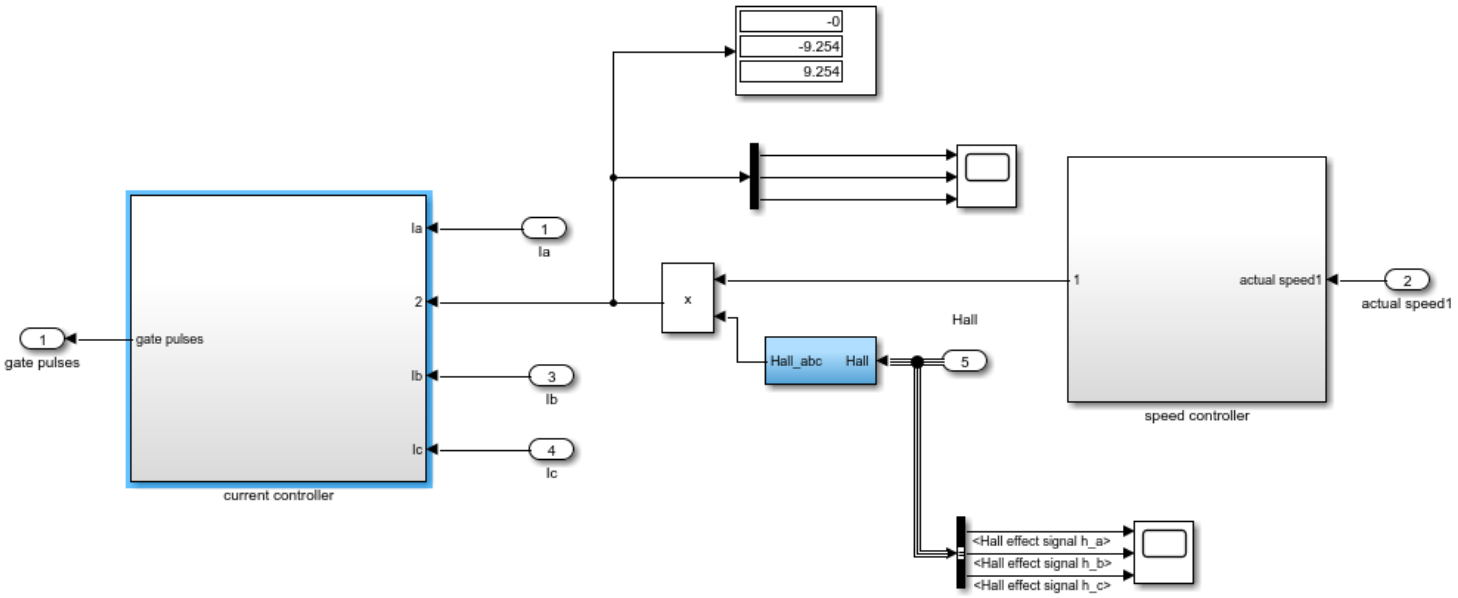


Figure 3.3.9: The model inside the controller block created

From figure 3.3.9, we can see that inside the controller we have a speed controller that takes the input from the speed variable provided by the permanent magnet synchronous machine. There is also a hall sensor that takes as input the hall effect signals from the three hall sensors that are inside the BLDCM. Finally, there is the current controller that takes four inputs and has one output which is the gate pulse.

For the speed controller:

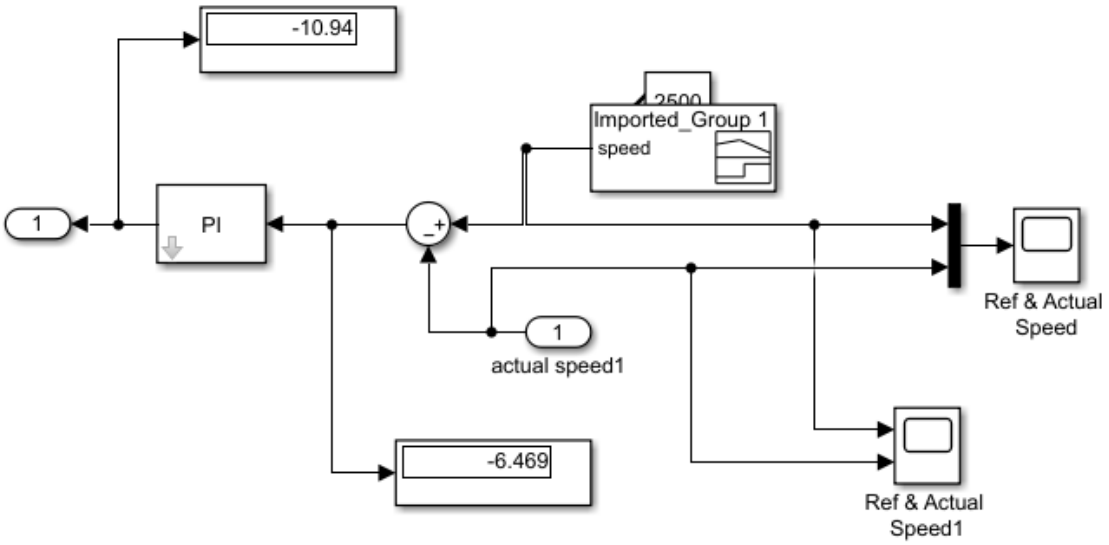


Figure 3.3.10: The speed controller model

To control the speed, we will use a PI controller which is one of the most ideal methods for a subtle transition between mechanical braking and electrical braking. The speed data used is imported from Simulink. The PI controller emits signals according to the error signal that is calculated in the sum block. The scope is used to compare the reference speed and the actual one.

The coefficients of the PI controller are shown in figure 3.3.11:

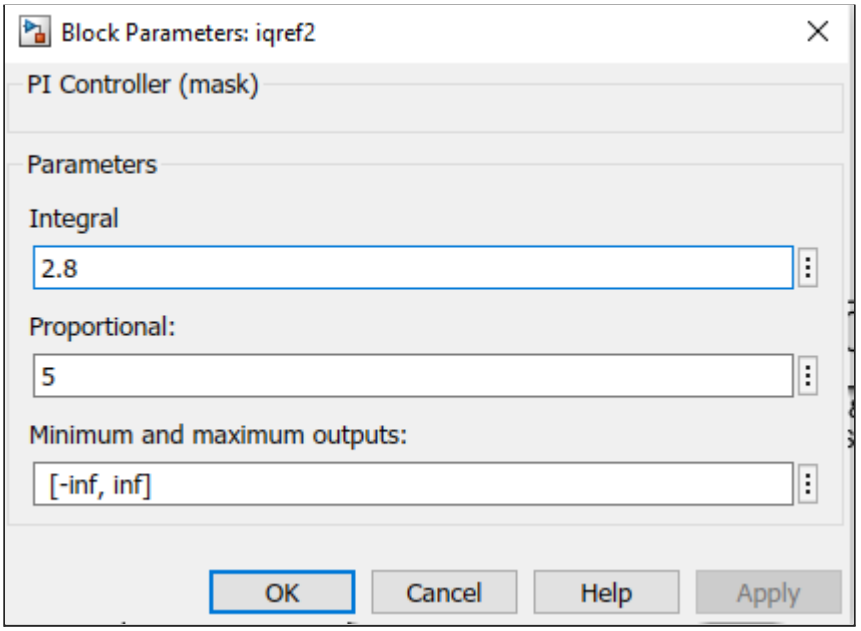


Figure 3.3.11: PI controller Coefficients

For the hall effect sensor, here is its circuit:

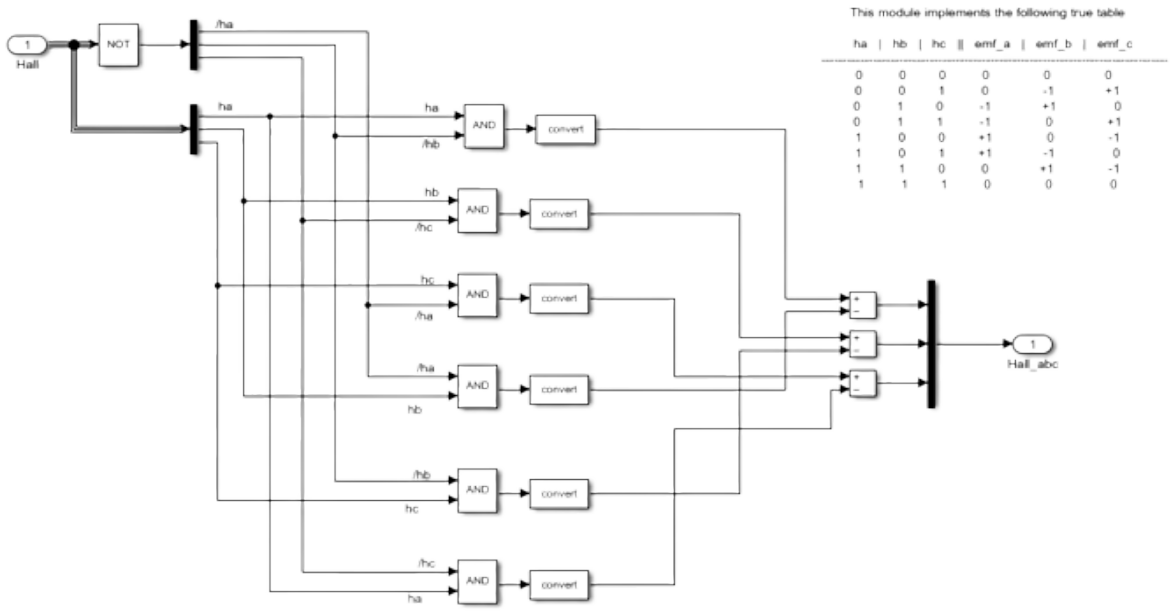


Figure 3.3.12: The Submodel inside the Hall Effect Sensor

The purpose of the hall sensor is to determine the rotor's position to detect the energized winding after the energizing sequence. [24]

For the current controller, here is its circuit:

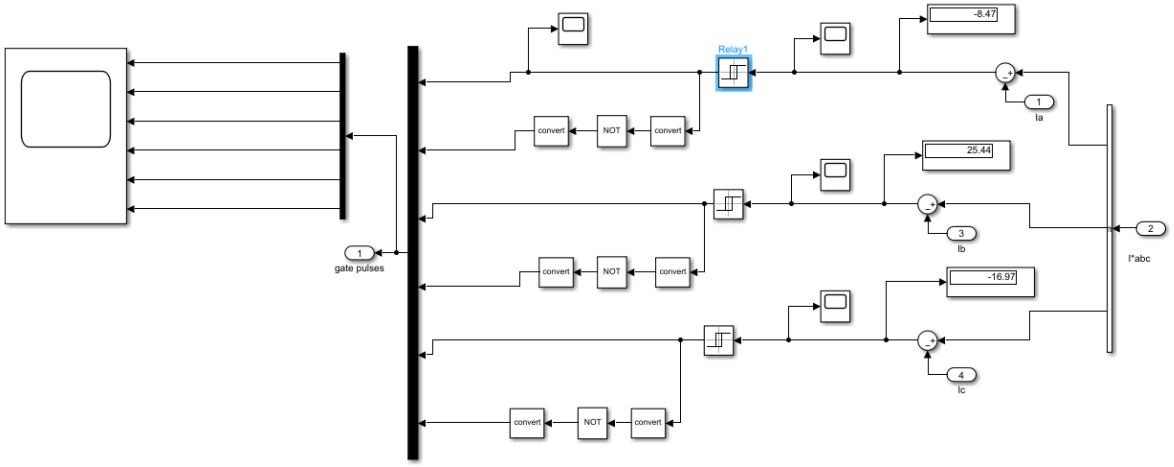


Figure 3.3.13: The Submodel inside the current controller

The output of the current controller will be the output of the whole controller block and it will be connected to the switching element MOSFET: the universal bridge block.

- ➢ Concerning the torque of the brake pedal, a Simulink function called signal builder was used to generate torque values.



Figure 3.3.14: Signal Builder Block

➔ Now after showing details of each block used, here is the final model:

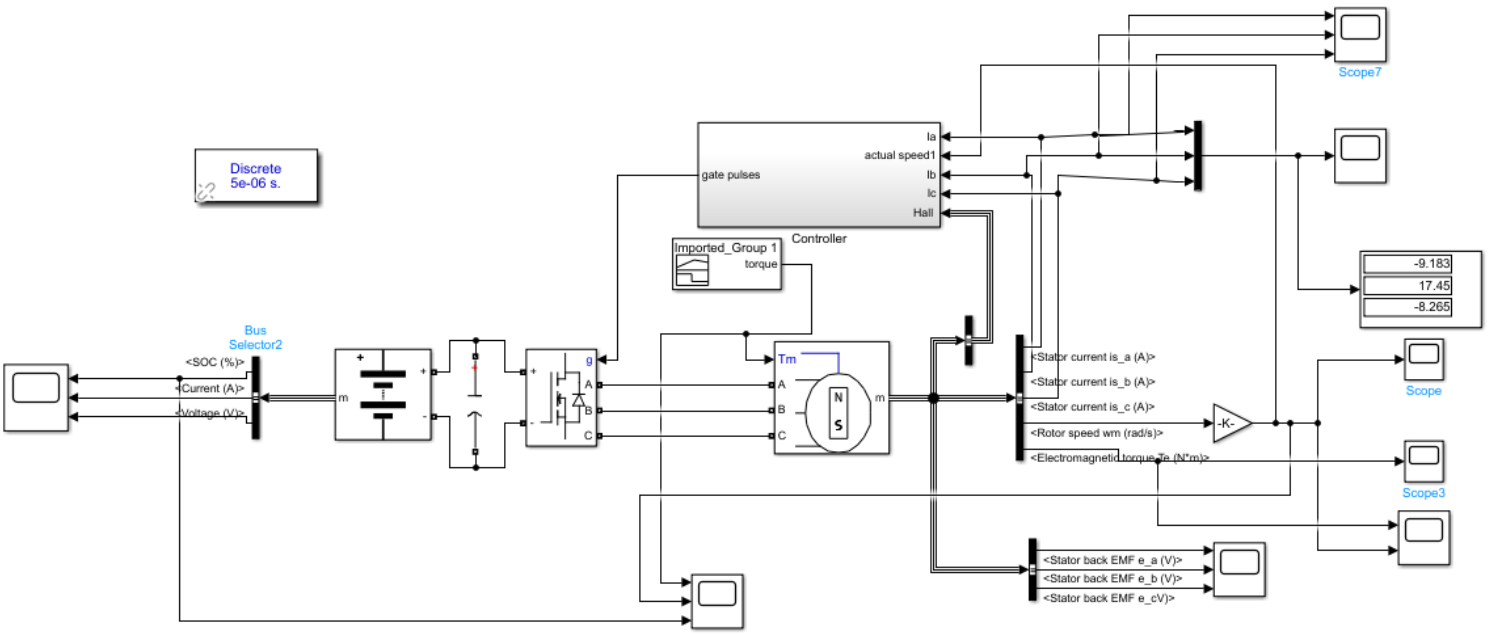


Figure 3.3.15: Simulation Circuit Model of the RBS using BLDCM

This figure represents the equivalent circuit and block diagram of the electric vehicle using BLDCM.

For the BLDCM to rotate, the phase should be switched in sequence; thus, the rotor's position should be recognized to determine this sequence. The position of the rotor is determined using the hall effect sensor that is connected to the m-ends of the motor.

Figure 3.3.12 depicts how the MOSFETS switch operates in order to detect the correct position of the current phase.

This data is used to switch the corresponding MOSFETS. This was accomplished by dividing the signals into comparison blocks. The signal was sent to the MOSFETs' gate (G) pin. For comparison, the AND blocks shown in were used.

The hall sensor provided input 1. The switching element MOSFET, the universal bridge block, was connected to the output of Out1. During operation, the NOT logic operator determines the passive hall sensor. This aforementioned set up is used to locate the rotor’s position using the stator pole as a reference.

4) Results & Discussion:

Blocks and data acquisition were shown for the whole system in the simulation part. In this section, the results of the scopes attached to the blocks used will be presented and discussed.

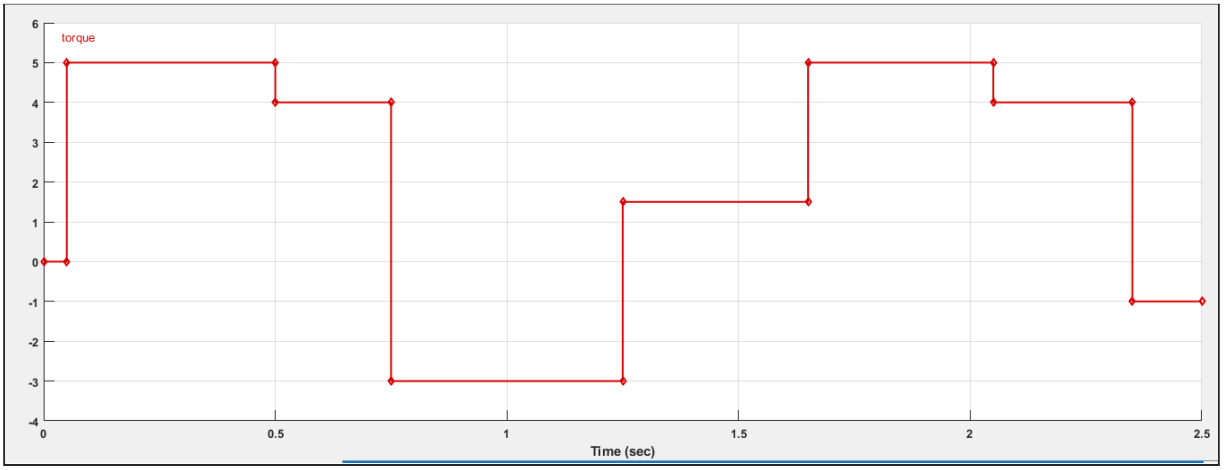


Figure 3.4.1: The torque generated by the signal builder

The figure represents the torque that enters the PMSM block. We can see that this machine operates in both modes: motor and generator. The sign of the toque governs the mode of operation (either motor mode when the torque is positive or generator mode when the torque is negative).

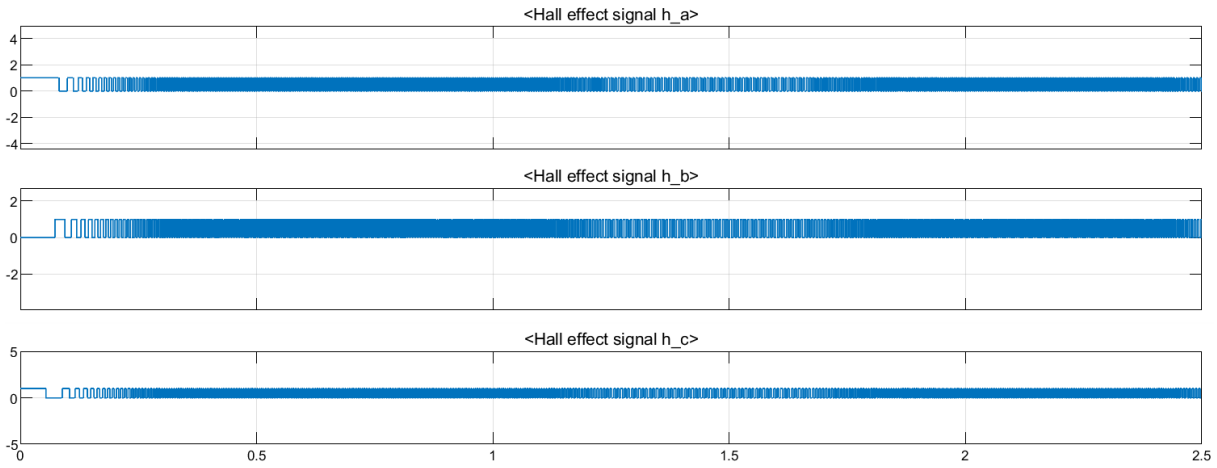


Figure 3.4.2: Simulation result of hall effect scope

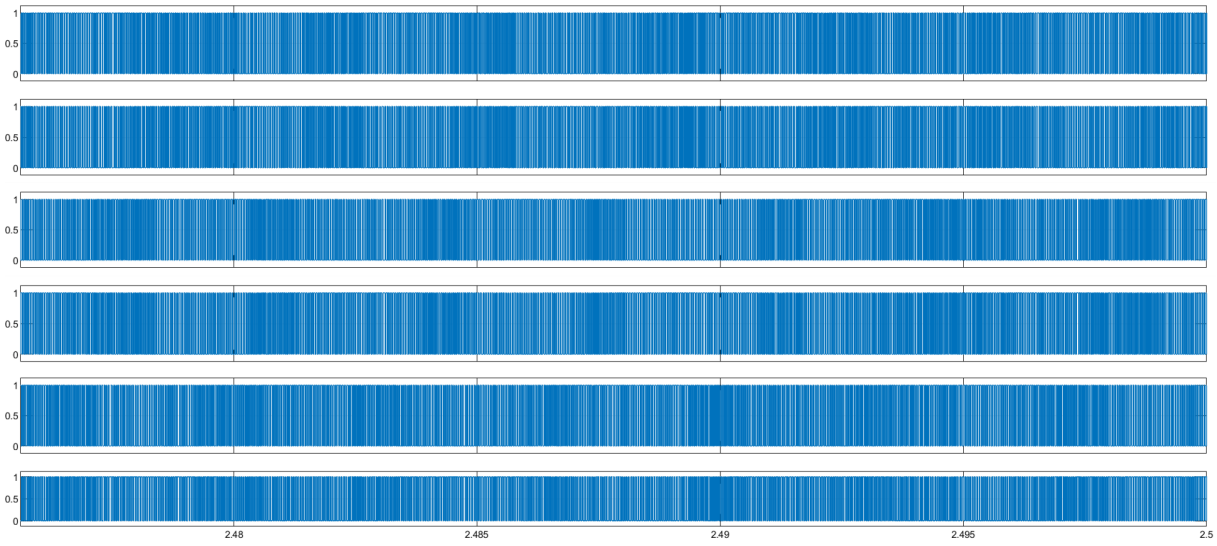


Figure 3.4.3: Simulation result of switches scope

The Hall Effect sensors and the switches are “OFF” when there is no current or not enough current in the coils. They only turn “ON”, when subjected to a magnetic field of sufficient strength and polarity which is shown in figures 3.4.2 and 3.4.3.

Let's check the values of the back emf voltages that are generated by the motor:

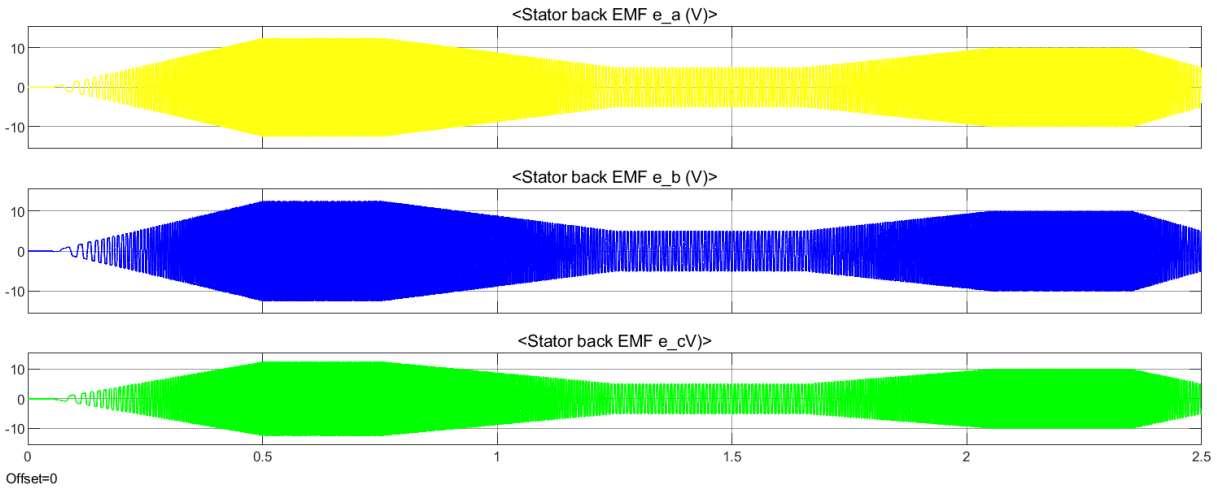


Figure 3.4.4: a. The Stator Back emf Signals

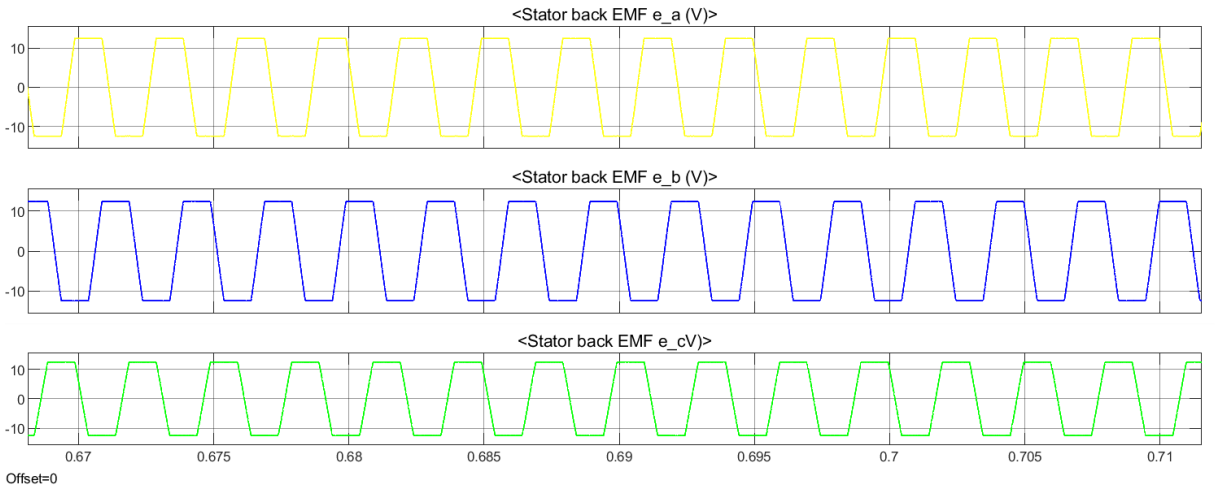


Figure 3.4.4: b.The Stator Back emf: Trapezoidal Signals

Figure 3.4.4. b zooms in figure 3.4.4.a.

Figure 3.4.4: b represents the back emf signals generated by each stator as we have phase a, phase b, and phase c. Since the signals are trapezoidal, then we can see that our motor is working properly.

Let's now check the speed of our system:

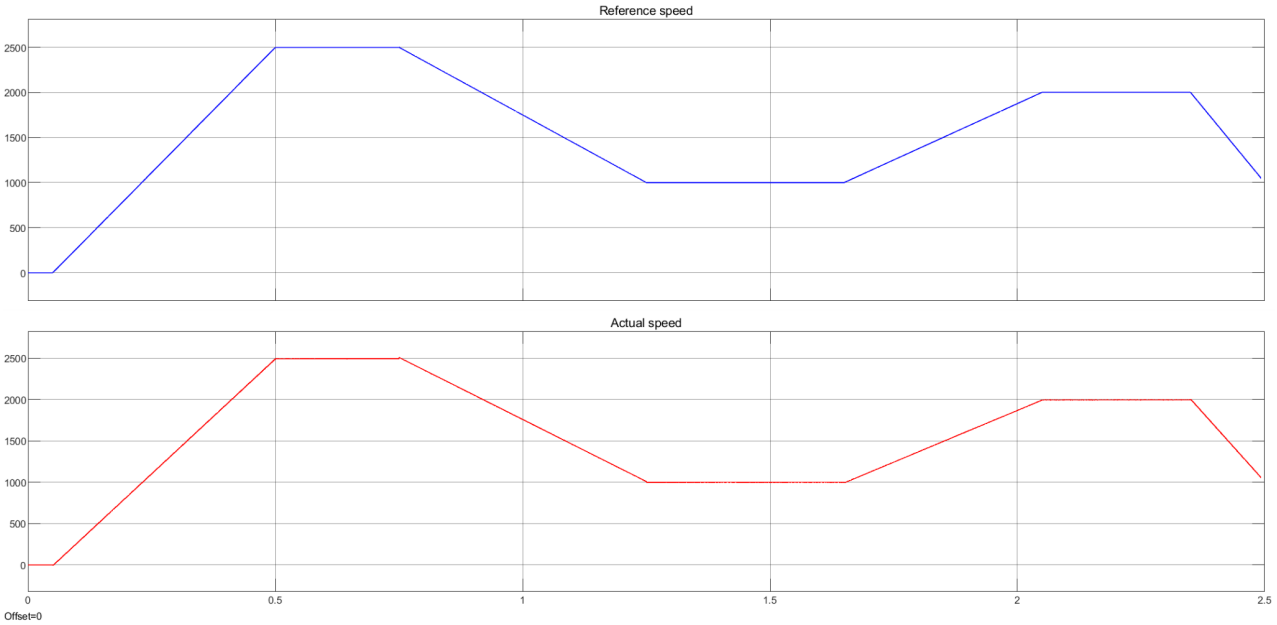


Figure 3.4.5: The reference speed Vs. The measured speed

From this figure, we can see that the speed is gradually achieved, hence, the motor along with the controller block are working correctly.

Let's check the current:

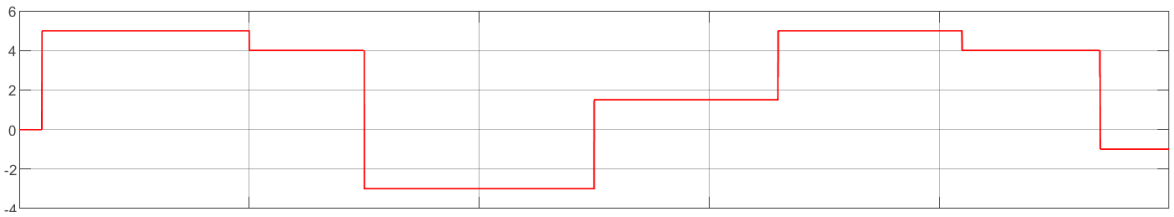


Figure 3.4.6: Current (A)

We have:

$$T_e = \frac{Power}{Speed} = \frac{V}{\omega} * I \quad (10)$$

So, the electromagnetic torque is proportional to the current which can be seen clearly when comparing figure 3.4.6 to figure 3.4.1.

Let's analyze the state of the battery:

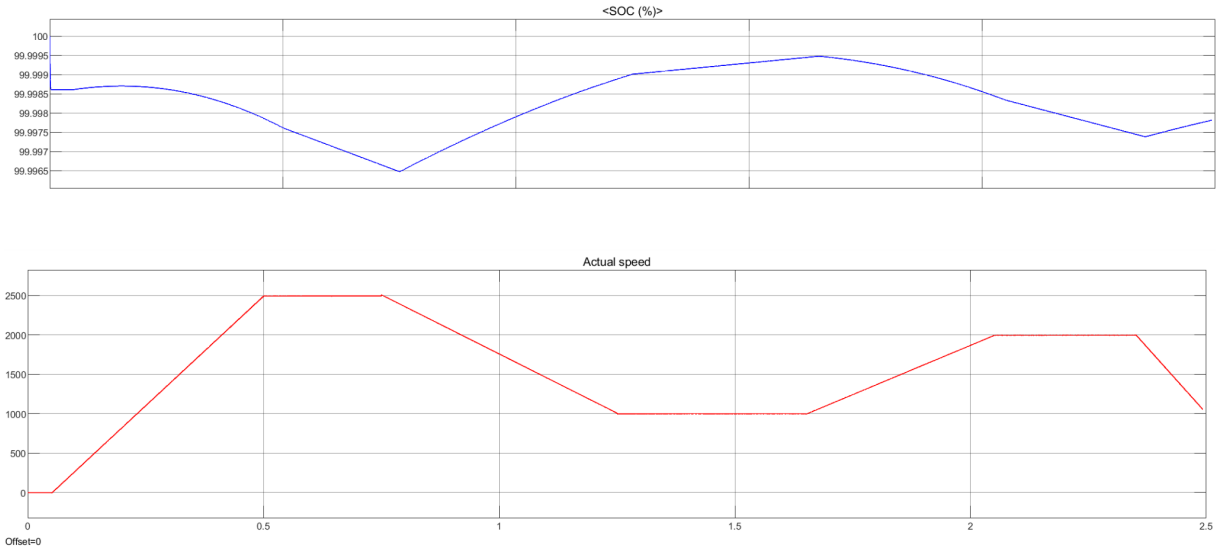


Figure 3.4.7: The SOC% Vs. The measured speed

SOC % represents the battery state of charge. As we see in the figure, the charge of the battery starts decreasing as the battery energizes the system, once the speed of the system starts

decreasing (braking mode) the charge of the battery increases fast. As we can see, regenerating power from braking occurs in bursts [24], the recovered energy can either be stored or used right away.

We can see from figure 3.4.7 that the system was able to recover 0.003% in only one brake operation:

At 0.75s, brake was performed:

% SOC recovered after brake = 99.9995 - 99.9965 = 0.003%

So, if the driver uses the brake pedal 50 times, he will be able to recover up to 1.5 % of %Soc.

➔ In this project, I explored an electric vehicle's regenerative braking system, which uses a Brushless DC motor to conduct the operation. To build this model, I used a battery, the Universal Bridge Block, BLDC motor, speed controller, PI controller, current controller, hall sensors along with other power electronics components.

The system was built in MATLAB Simulink in order to verify its performance and to test the recovery system. As a result, the RBS model was able to recover 0.003% of the lost energy in each brake operation. This amount of energy will be significant since brakes are used hundreds of times a day.

Future Work:

This regenerative braking system was built with the aim of increasing the efficiency of the system so that our system can recover the maximum energy possible.

Here are my next steps:

- Control the motor using PWM.
- Research neural network sliding mode controllers to improve and recover more energy.
- Work on a control loop to increase efficiency.
- Focus on the battery lifetime.

STEEPLE ANALYSIS:

Socially:

The social impact of this project is rather positive since I am opting for green technologies to recover energy. Besides, it targets all vehicle users worldwide. This project encourages research in the field in order to increase the efficiency of already existing regenerative braking systems and reduce the need for electricity by using renewable sources.

Technologically:

The main purpose of this capstone work is to model a regenerative braking system based on a BLDC motor that will be used in an electric vehicle with the aim of improving its efficiency later on. This project is exploring technology to see how I can use technology to serve the client and the environment as well.

Economically:

This project aims to reduce the user's need for electricity, hence, the user will save up money. Though initial costs are somewhat high, the cost while calculating over a period of time will be reduced which shows the impact of the system.

Environmentally:

This project is eco-friendly since it reduces wear and tear and improves the life of the components. Most importantly, it resumes noise while decelerating. This project uses a renewable energy source to charge the battery such that it recovers the kinetic energy lost as heat and transforms it into electricity.

Politically:

This project will encourage foreign companies specialized in producing electric vehicles to open up new branches in the kingdom, hence, encouraging the local industry. Besides, it will encourage partnerships in the field with other countries.

Legal Factors:

The system is expected to impose the same safety standards that are currently used by the automotive industry.

Ethical Factors:

The RBS is expected to comply with the ethical standards imposed by the automotive industry.

Conclusion:

When a vehicle is in motion, it has a large amount of kinetic energy, and when the brakes are used to slow it down, all that kinetic energy is simply lost to the environment as heat.

This heat can be transformed into useful energy that can charge the battery using a regenerative braking system. This braking system still depends on different factors. However, it has many benefits including increased driving range, enhanced braking effectiveness, reduced brake wear, and so on. In this paper, a regenerative braking system that recovers up to 0.003% per brake operation was built using a BLDC motor and was simulated in the Simulink environment to assess how the recovery system operates. This project is just the first step to improving the efficiency of a BLDCM based RBS.

REFERENCES:

[1] Joseph Godfrey, A., & Sankaranarayanan, V. (2018). A new electric braking system with energy regeneration for a BLDC Motor Driven Electric Vehicle. Engineering Science and Technology, an International Journal, 21(4), 704–713.

[2] Regenerative braking. Regenerative braking - Energy Education. (n.d.). Retrieved April 17, 2022, from https://energyeducation.ca/encyclopedia/Regenerative_braking.

[3] R. Kapoor and M. Parveen, "Comparative Study on Various KERS," Proceedings of the World Congress on Engineering, vol. 3, Jul. 2013.

[4] J. Larminie, Electric vehicle technology explained. Chichester, West Sussex, United Kingdom: Wiley a John Wiley & Sons Ltd. Publication, 2012.

[5] Lamichhane, M., Bhetal, N., Yonjan, S., & Y, V. S. R. O. (2020, January 1). (PDF) design and fabrication of Regenerative Braking System. ResearchGate. Retrieved April 17, 2022, from https://www.researchgate.net/publication/341379119_Design_and_Fabrication_of_Regenerative_Braking_System

[6] Design, fabrication and testing of regenerative braking ... (n.d.). Retrieved April 17, 2022, from https://www.irjet.net/archives/V3/i5/IRJET-V3I5386.pdf

[7] S.J Clegg, "A Review of Regenerative Braking Systems", White Rose Research Online", Apr 1996. [5] M. Goodman, Regenerative Braking.

[8] J. Ewbank, "How Regenerative Braking Works," Engineering.com, Sep 2018. [Online]. Available: https://www.engineering.com/ElectronicsDesign/ElectronicsDesignArticles/ArticleID/17625/How-Regenerative-Braking-Works.aspx.

[9] Chueprasert, W., & Phaoharuhansa, D. (2020). Study of regenerative braking system and brake force using pulse width module. *MATEC Web of Conferences*, *306*, 01004. https://doi.org/10.1051/matecconf/202030601004

[10] S.R. Cikanek and K.E. Bailey. Regenerative braking system for a hybrid vehicle. American Control Conference, 2002.

[11] Sindhuja, V., Ranjitham, G., (2014). Regenerative braking system of electric vehicle driven by BLDC motor using neuro fuzzy and PID. International Journal of Innovative Research in Science, Engineering and Technology, 3(12), 17847-17854.

[12] Kumar, E. A. (2022, February 26). What is a brake? Types of braking system [notes & PDF]. THEMECHANICALENGINEERING.COM. Retrieved April 17, 2022, from https://themechanicalengineering.com/braking-system/#2_Hydraulic_Braking_System

[13] Energy storage - mechanical systems - springs. (n.d.). Retrieved April 17, 2022, from https://technologystudent.com/energy1/storage1.html

[14]What are the different types of brake systems and how do they work - autodoc. (n.d.). Retrieved April 17, 2022, from https://club.autodoc.co.uk/magazin/what-are-the-different-types-of-brake-systems-and-how-do-they-work .

[15] Els, P. (2019, February 5). Regenerative braking in an electrified/autonomous landscape. Automotive IQ. Retrieved April 17, 2022, from https://www.automotive-iq.com/autonomous-drive/articles/regenerative-braking-in-an-electrifiedautonomous-landscape

[16]What is the regenerative braking system? Electric vehicles. (n.d.). Retrieved April 17, 2022, from https://www.alke.com/brake-energy-recovery

[17]What are brushless DC Motors? Renesas. (n.d.). Retrieved April 17, 2022, from https://www.renesas.com/us/en/support/engineer-school/brushless-dc-motor-01-overview

[18] Technology, E., Says:, J., Says:, R. P., Says:, M. I., Says:, M., says:, H., says:, K. Y. T., says:, T. F., & Says:, S. P. (2020, October 14). What is a brushless DC Motor (BLDC)? Construction & Working. ELECTRICAL TECHNOLOGY. Retrieved April 17, 2022, from https://www.electricaltechnology.org/2016/05/bldc-brushless-dc-motor-construction-working-principle.html

[19] Six pulse bridge converter: Performance characteristics. EEEGUIDE.COM. (2022, April 6). Retrieved April 17, 2022, from https://www.eeeguide.com/six-pulse-bridge-converter/

[20] Collins, D. (n.d.). What's the relationship between current and DC motor output torque? Motion Control Tips. Retrieved April 17, 2022, from https://www.motioncontroltips.com/faq-whats-the-relationship-between-current-and-dc-motor-output-torque/

[21] Sindhuja, V., Ranjitham, G., (2014). Regenerative braking system of electric vehicle driven by BLDC motor using neuro fuzzy and PID. International Journal of Innovative Research in Science, Engineering and Technology, 3(12), 17847-17854.

[22] Kivanc, O. C., & Ustun, O. (2021, January 24). Investigation of regenerative braking performance of Brushless Direct Current Machine Drive System. MDPI. Retrieved April 17, 2022, from https://www.mdpi.com/2076-3417/11/3/1029

[23] Gamazo-Real, J. C., Vázquez-Sánchez, E., & Gómez-Gil, J. (2010). Position and speed control of brushless DC motors using sensorless techniques and application trends. Sensors (Basel, Switzerland). Retrieved April 17, 2022.

[24] Godfrey A J and Sankaranarayanan V 2018 Engg. Sci. Technol. Int. J.

Sayed Nashit, Sufiyan Adhikari, Shaikh Farhan, Srivastava Avinash and Amruta Gambhire, 'Design, Fabrication and Testing of Regenerative Braking Test Rig for BLDC Motor', 2016.

[25] J, M., & M, A. (2019). A review on regenerative braking methodology in Electric Vehicle. International Journal of Psychosocial Rehabilitation, 23(4), 220–226. https://doi.org/10.37200/ijpr/v23i4/pr190180

[26] Robinson, M. (2018, June 18). The best-sounding Le mans prototypes use a V8 from an SUV. Car Throttle. Retrieved April 18, 2022, from https://www.carthrottle.com/post/the-best-sounding-prototypes-at-le-mans-dont-take-part-in-the-race/